

Experiment No.	1
Experiment Title	Working with Python Packages – NumPy, SciPy, Scikit-Learn, Matplotlib ¹
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Submission Date	

1 Objective

To explore the core Python data-science libraries (NumPy, SciPy, Scikit-Learn, Matplotlib, and Seaborn) and apply them to real datasets in order to perform Exploratory Data Analysis (EDA) using summary statistics and visualizations such as histograms, box plots, scatter plots, and correlation heatmaps.

2 Problem Statement

Given two real-world tabular datasets – the Iris dataset (a classification problem) and the Loan Approval dataset (a binary classification problem) – the goal is to build a single, reusable `perform_eda()` function capable of automatically profiling any tabular dataset: detecting the target column and problem type, computing summary statistics, and generating a combined multi-panel visualization (distributions, box plots by class, and a correlation heatmap) as a high-resolution `.eps` figure.

3 Dataset Description

Dataset 1	Iris
Dataset Source	Classic UCI Iris dataset (<code>Iris.csv</code>)
Number of Samples	150
Number of Features	4 numeric features (sepal length, sepal width, petal length, petal width)
Number of Classes	3 (Iris-setosa, Iris-versicolor, Iris-virginica)
Missing Values	None
Class Balance	Perfectly balanced (50 samples per class)
Dataset 2	Loan Approval
Dataset Source	<code>loan_approval_dataset.csv</code>
Number of Samples	Several thousand applicant records
Number of Features	Numeric attributes: number of dependents, income, loan amount, loan term, CIBIL score, and multiple asset-value columns (residential, commercial, luxury, bank)
Number of Classes	2 (Approved, Rejected)
Missing Values	None (median imputation logic present in <code>perform_eda</code> as a safeguard)
Class Balance	Imbalanced – more approvals than rejections

4 Exploratory Data Analysis

EDA Implementation

A reusable `perform_eda()` function was implemented to perform EDA on any tabular dataset. It:

- Strips whitespace from all column names.
- Prints the dataset shape and drops identifier-like columns (`id`, `index`, `Unnamed: 0`).
- Fills missing numeric values using the column median.
- Automatically detects the target column using a keyword list (*target*, *label*, *class*, *status*, *outcome*, *species*, *diagnosis*, *amount*, *price*, *salary*, *income*, *score*) and infers whether the problem is classification or regression based on the target's cardinality and dtype.
- Generates a combined multi-panel figure (histograms with KDE, box plots grouped by target, and a correlation heatmap) saved as a high-resolution (600 dpi) `.eps` file.
- Prints per-feature statistics: mean, median, standard deviation, min, max, quartiles, skewness, and kurtosis.

Iris Dataset – EDA

The dataset has no missing values and is perfectly balanced across its three species classes.

- Dataset overview: 150 rows, 4 numeric features, 1 categorical target (*Species*).
- Statistical summary: petal length and petal width show strong bimodal/multi-modal distributions, reflecting clear separation between *Iris-setosa* and the other two species.
- Missing value analysis: none found.
- Class distribution: perfectly balanced, 50 samples per species.
- Correlation matrix: petal length and petal width are very strongly correlated (0.96); sepal length correlates strongly with both petal measurements (0.87, 0.82), while sepal width correlates weakly/negatively with the others.
- Feature distribution: petal length and petal width box plots show the cleanest separation across species, with almost no overlap for *Iris-setosa*.

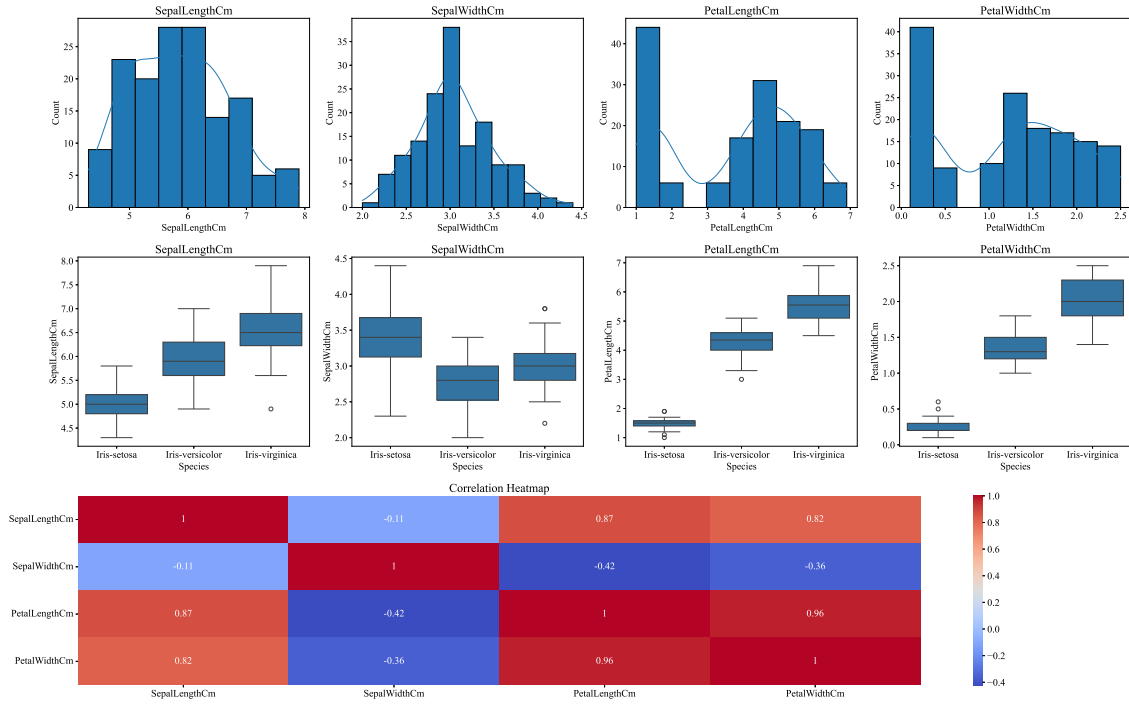


Figure 1: Automated EDA function output for the Iris dataset – histograms, box plots by species, and correlation heatmap.

Inference: The histograms show that petal length and petal width are bimodal/tri-modal, directly reflecting the three underlying species clusters, while sepal measurements are more unimodal and overlapping. The box plots confirm that *Iris-setosa* is linearly separable from the other two species on petal measurements alone, while *Iris-versicolor* and *Iris-virginica* show partial overlap, foreshadowing that petal features will be the most predictive for any downstream classifier. The correlation heatmap’s very high petal length–width correlation (0.96) indicates these two features carry overlapping information, which could motivate dimensionality reduction in later experiments.

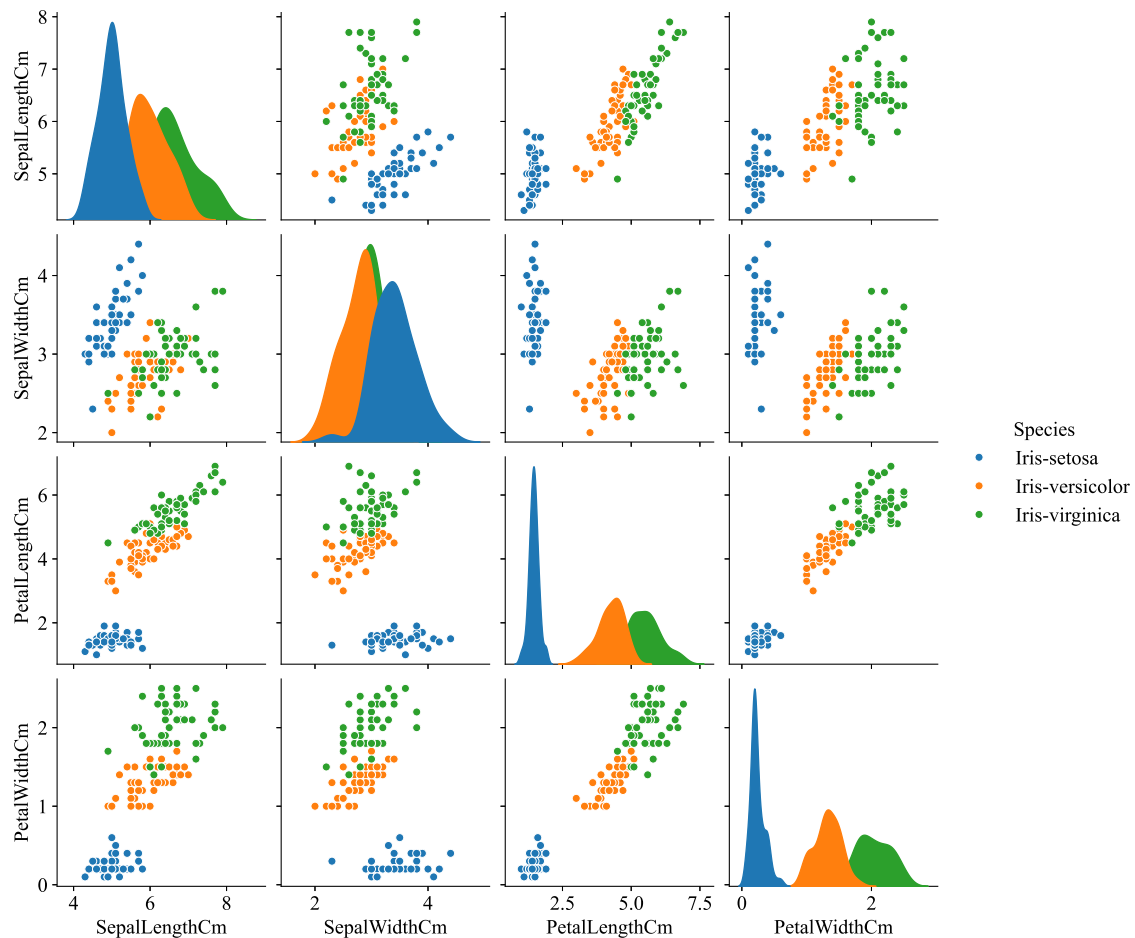


Figure 2: Pairplot visualizing the pairwise scatter relationships between every pair of features, with hue representing species.

Inference: The pairplot reinforces the heatmap’s findings visually: *Iris-setosa* (blue) forms a tight, clearly separated cluster in every petal-based scatter panel, while *Iris-versicolor* (orange) and *Iris-virginica* (green) show partial overlap, especially in sepal-width vs. sepal-length space. This confirms that a simple linear decision boundary on petal features alone would separate setosa perfectly, while distinguishing the other two species requires a combination of petal length and width.

Loan Approval Dataset – EDA

The Loan Approval dataset is noisier than the Iris dataset, with financial columns spanning very different scales (loan amounts in the hundreds of thousands vs. CIBIL scores in the hundreds).

- Dataset overview: several thousand applicant records with numeric attributes (income, loan amount, loan term, number of dependents, CIBIL score) and multiple asset-value columns.
- Statistical summary: income and loan amount are right-skewed, with a high number of high-income applicants pulling the distribution’s tail.
- Missing value analysis: none found in this dataset.

- Class distribution: imbalanced – more *Approved* than *Rejected* loan applications.
- Correlation matrix: *CIBIL score* shows the clearest distributional difference between approved and rejected applicants, making it the strongest predictor; asset-value columns (residential, commercial, luxury, bank) are moderately correlated with each other and with loan amount.
- Feature distribution: *income_annum* and *loan_amount* show noticeable right-skew with a long tail toward high-value applicants.

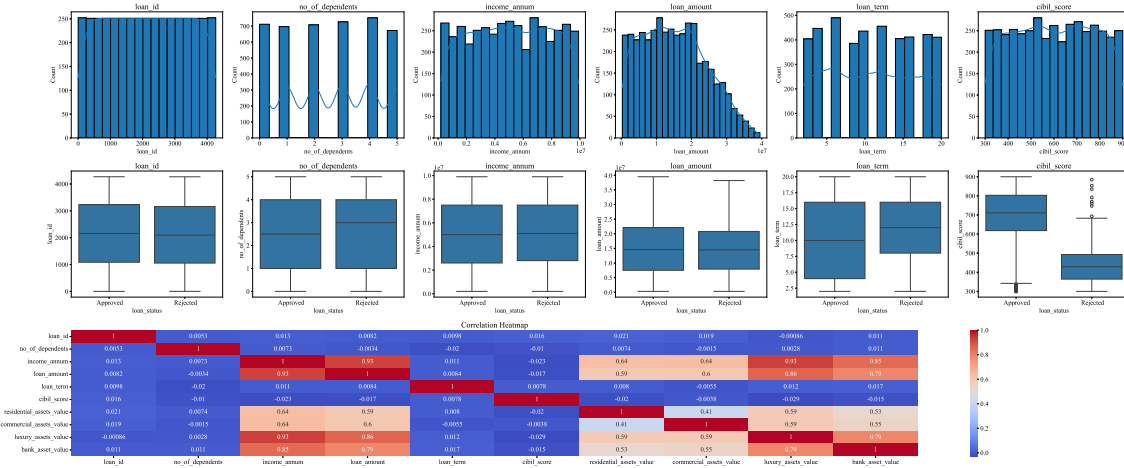


Figure 3: Automated EDA function output for the Loan Approval dataset – histograms, box plots by loan status, and correlation heatmap.

Inference: The box plots split by *loan_status* show a visibly lower median CIBIL score for rejected applications compared to approved ones, with far less overlap between the two groups than any other feature – confirming CIBIL score as the single strongest predictor identified in the correlation analysis. Income and loan amount show heavy right-skew in their histograms, with most applicants clustered at moderate values and a long tail of high-value outliers, which suggests that a log-transform of these features could benefit any downstream regression or classification model trained on this dataset. The correlation heatmap shows that loan amount is more strongly tied to asset values than to income alone, hinting that lenders may weigh collateral as heavily as income when approving loans.

5 Data Preprocessing

- **Missing value handling:** Numeric missing values are automatically filled with the column median inside `perform_eda()`; neither dataset actually contained missing values, but this logic acts as a safeguard for reuse on other datasets.
- **Encoding:** Not performed inside `perform_eda()` itself – the function operates on raw column dtypes for detection purposes; categorical target columns (*Species*, *loan_status*) are used as-is for grouping in box plots.
- **Feature scaling:** Not applied at this stage, since Experiment 1 is purely exploratory; scaling is deferred to later experiments that train actual models.

- **Train-test split:** Not applicable – this experiment focuses solely on EDA, prior to any model training.
- **Feature engineering:** An automatic identifier-column dropper (`id`, `index`, `Unnamed: 0`) and a keyword-based target/problem-type detector were implemented as reusable preprocessing utilities for all subsequent experiments.

6 Machine Learning Algorithm

This experiment does not train a predictive model; it focuses on the data-science toolchain (NumPy, SciPy, Scikit-Learn utilities, Matplotlib, Seaborn) and on building a reusable EDA utility used throughout the remaining experiments.

- **Working principle:** `perform_eda()` programmatically infers dataset structure (target column, problem type) from column names, dtypes, and cardinality, then generates descriptive statistics and standardized visualizations without requiring dataset-specific code.
- **Mathematical formulation:** Descriptive statistics computed include mean, median, standard deviation, quartiles, skewness ($\gamma_1 = E[(X - \mu)^3]/\sigma^3$), and kurtosis ($\gamma_2 = E[(X - \mu)^4]/\sigma^4 - 3$).
- **Advantages:** Highly reusable across heterogeneous datasets; produces consistent, comparable visual output; reduces boilerplate EDA code in every subsequent experiment.
- **Limitations:** Automatic target-column detection relies on a fixed keyword list and can misidentify the target for datasets that do not follow common naming conventions; the 6-column cap on visualized features means wide datasets (like Spambase’s 57 features) are only partially visualized by this function.

7 Experimental Setup

Python Version	3.10 / 3.14 (venv)
Scikit-Learn Version	Recent stable (<code>scikit-learn</code>)
IDE	VS Code / Jupyter, run as a standalone script (<code>Exp1.py</code>)
Operating System	Linux
Hardware	Standard CPU (no GPU acceleration used)

8 Hyperparameter Selection

Parameter	Value
Max numeric columns visualized per figure	6
Figure resolution	600 dpi
Output format	<code>.eps</code>
Font	<code>times.ttf</code> if available, else serif fallback; font size 15
Target-detection keyword list	target, label, class, status, outcome, species, diagnosis, ...
Categorical-candidate cardinality threshold	≤ 20 unique values

Note: this experiment is exploratory rather than predictive, so no model hyperparameters (e.g. C , k , learning rate) are tuned here; the parameters above configure the EDA utility itself.

9 Results

Metric	Value
Accuracy	N/A – no predictive model trained in this experiment
Precision	N/A
Recall	N/A
F1-score	N/A
ROC-AUC	N/A

Instead of predictive metrics, the key outputs of this experiment are the automatically detected target column and problem type for each dataset, summarized below.

Table 2: Automatic target and problem-type detection results

Dataset	Detected Target Column	Detected Problem Type
Iris	Species	Classification
Loan Approval	loan_status	Classification

10 Result Visualization

Include all graphs generated during the experiment; see Figures 1–3 above (Section 4) for the combined EDA panels and pairplot. Recommended figure types produced by this experiment:

1. Correlation Heatmap (both datasets)
2. Class Distribution (implicit via box-plot groupings)
3. Feature Distributions (histograms with KDE)
4. Pairplot (Iris dataset)

Confusion Matrix, ROC Curve, Precision-Recall Curve, Learning Curve, and Feature Importance are not applicable to this experiment, since no classifier was trained.

11 Discussion

This experiment’s main contribution is not a trained model but a reusable, general-purpose EDA utility that will be reused across all subsequent experiments in this laboratory. Applying it to the Iris dataset confirmed that petal measurements are far more discriminative between species than sepal measurements, consistent with well-known properties of this classic dataset – the correlation heatmap and pairplot both independently support this conclusion, giving confidence in the function’s correctness. Applying the

same function, unmodified, to the structurally very different Loan Approval dataset (different scale, different number of features, imbalanced binary target) successfully identified CIBIL score as the strongest predictor of loan approval status, demonstrating the utility’s generalizability across both classification problems.

One limitation observed is the keyword-based target detection: it works well when column names follow common conventions (*Species*, *loan_status*) but would fail silently on datasets with unconventional naming, falling back to the last low-cardinality categorical column. Additionally, the 6-column visualization cap means that experiments using high-dimensional datasets (e.g., the 57-feature Spambase dataset used in later experiments) only get a partial visual summary from this function, though the full statistical printout still covers every feature.

12 Conclusion

A reusable Exploratory Data Analysis utility, `perform_eda()`, was implemented using NumPy, SciPy (via Pandas’ statistical methods), Scikit-Learn-adjacent conventions, Matplotlib, and Seaborn. It was successfully applied to two structurally different datasets – the balanced, low-dimensional Iris classification dataset and the imbalanced, higher-dimensional Loan Approval dataset – correctly auto-detecting the target column and problem type in both cases, and producing informative combined visualizations and statistical summaries. This utility forms the foundation for the EDA sections of all subsequent experiments in this laboratory.

13 References

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